ENGINEERING ECONOMY

LECTURE 4

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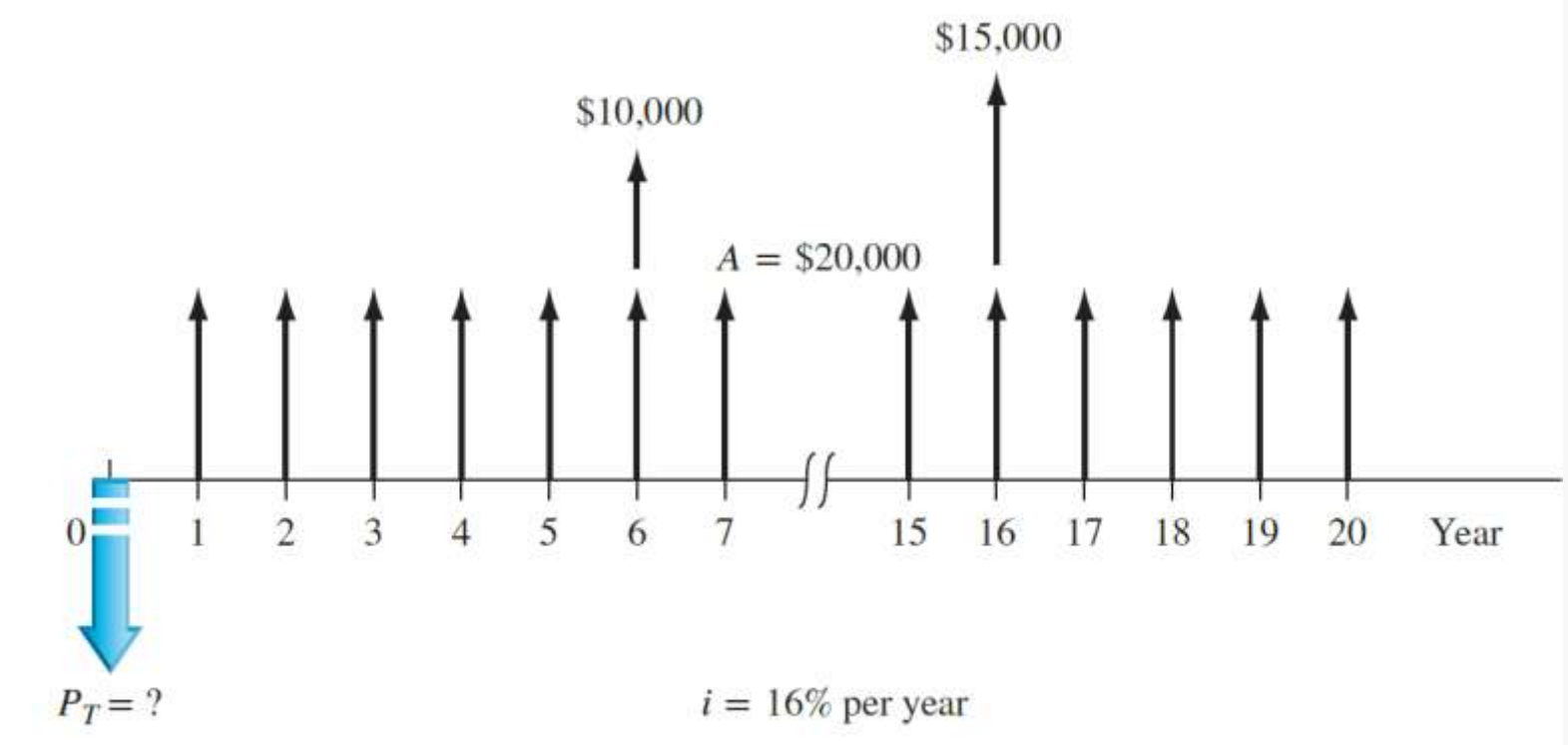
Calculations Involving Uniform Series and Randomly Placed Single Amounts

- When a cash flow includes both a uniform series and randomly placed single amounts, the procedures of previous lectures are applied to the uniform series and the single-amount formulas are applied to the one-time cash flows.

EXAMPLE

- An engineering company in Wyoming that owns 50 hectares of valuable land has decided to lease the mineral rights to a mining company. The primary objective is to obtain long-term income to finance ongoing projects 6 and 16 years from the present time. The engineering company makes a proposal to the mining company that it pay $20,000 per year for 20 years beginning 1 year from now, plus $10,000 six years from now and $15,000 sixteen years from now. If the mining company wants to pay off its lease immediately, how much should it pay now if the investment is to make 16% per year?

- Find the present worth of the 20-year uniform series and add it to the present worth of the two one-time amounts to determine P_T'.



$$P_T = 20{,}000(P/A,16\%,20) + 10{,}000(P/F,16\%,6) + 15{,}000(P/F,16\%,16)$$

$$= \$124{,}075$$

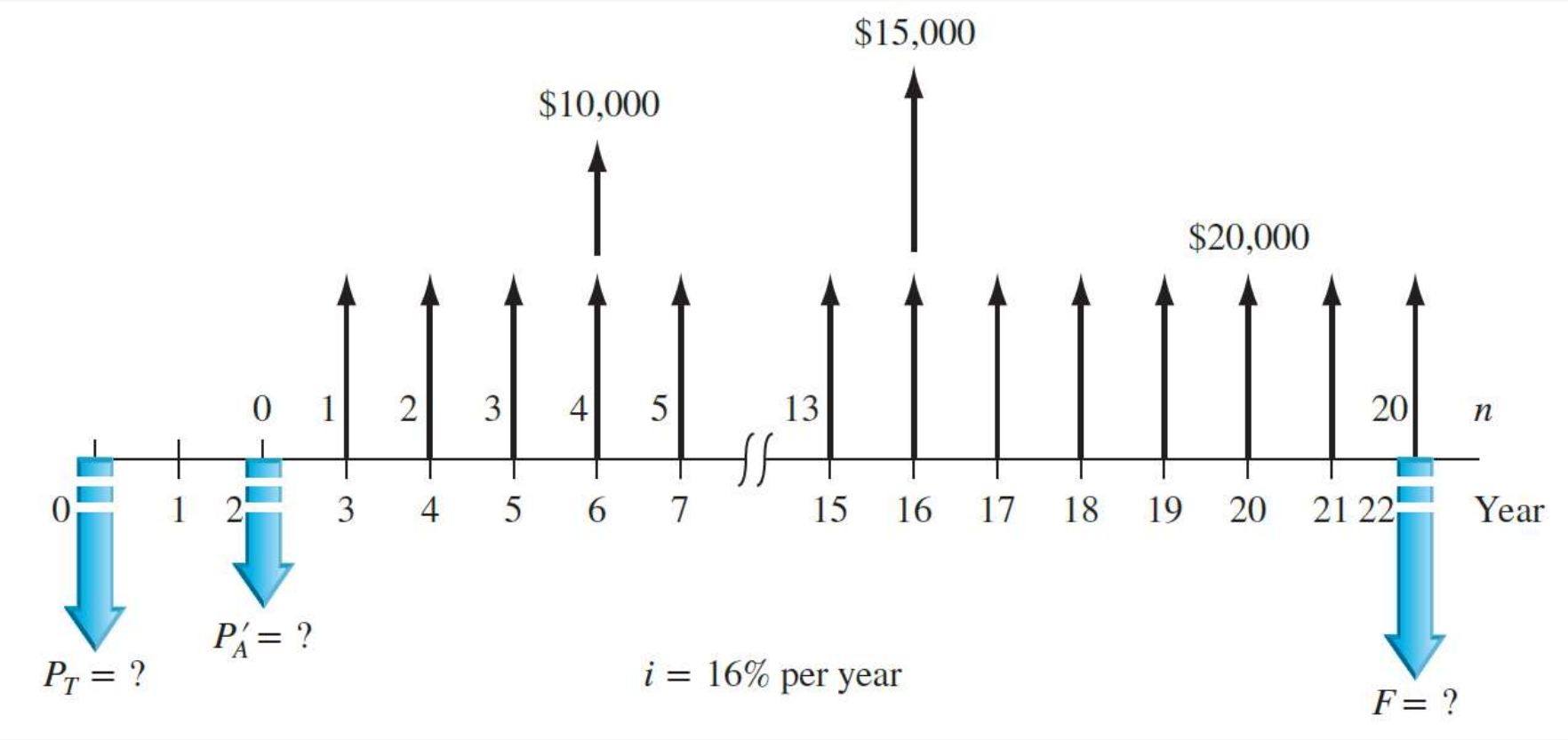
- When you calculate the *A* value for a cash flow series that includes randomly placed single amounts and uniform series, **first convert everything to a present worth or a future worth.**
- Then you obtain the *A* value by multiplying *P* or *F* by the appropriate *A/P* or *A/F* factor.

- *A* value by multiplying *P* or *F* by the appropriate *A/P* or *A/F* factor.

EXAMPLE:

- A design-build-operate engineering company in Texas that owns a sizable amount of land plans to lease the drilling rights (oil and gas only) to a mining and exploration company. The contract calls for the mining company to pay $20,000 per year for 20 years beginning 3 years from now (i.e., beginning at the end of year 3 and continuing through year 22) plus $10,000 six years from now and $15,000 sixteen years from now. Utilize engineering economy relations to determine the five *equivalent values* listed below at 16% per year.

    1. Total present worth P_T in year 0
    2. Future worth F in year 22
    3. Annual series over all 22 years
    4. Annual series over the first 10 years
    5. Annual series over the last 12 years



- **1. P_T in year 0:** First determine P'_A of the series in year 2. Then P_T is the sum of three P values: the series present worth value moved back to $t = 0$ with the P/F factor, and the two P values at $t = 0$ for the two single amounts in years 6 and 16.

$$P'_A = 20{,}000(P/A,16\%,20)$$

$$P_T = P'_A(P/F,16\%,2) + 10{,}000(P/F,16\%,6) + 15{,}000(P/F,16\%,16)$$

$$= 20{,}000(P/A,16\%,20)(P/F,16\%,2) + 10{,}000(P/F,16\%,6) + 15{,}000(P/F,16\%,16)$$

$$= \$93{,}625$$

- **2. *F* in year 22:** To determine *F* in year 22 from the original cash flows, find *F* for the 20-year series and add the two *F* values for the two single amounts. Be sure to carefully determine the *n* values for the single amounts: $n = 22 - 6 = 16$ for the \$10,000 amount and $n = 22 - 16 = 6$ for the \$15,000 amount.

$$F = 20{,}000(F/A,16\%,20) + 10{,}000(F/P,16\%,16) + 15{,}000(F/P,16\%,6)$$

$$= \$2{,}451{,}626$$

- **3. *A* over 22 years:** Multiply P_T = \$93,625 by the *A*/*P* factor to determine an equivalent 22-year *A* series, referred to as A_{1-22} here.

$$A_{1-22} = P_T(A/P,16\%,22) = 93{,}625(0.16635) = \$15{,}575$$

- OR

$$A_{1-22} = F(A/F,16\%,22) = \$15{,}575$$

- **4. *A* over years 1 to 10:** The equivalent *A* series is calculated for a number of years different from that covered by the original cash flows.
- This occurs when a defined *study period* or *planning horizon* is preset for the analysis.
- To determine the equivalent *A* series for years 1 through 10 only (call it A_{1-10}), the P_T *value must be used* with the *A/P* factor for $n = 10$.

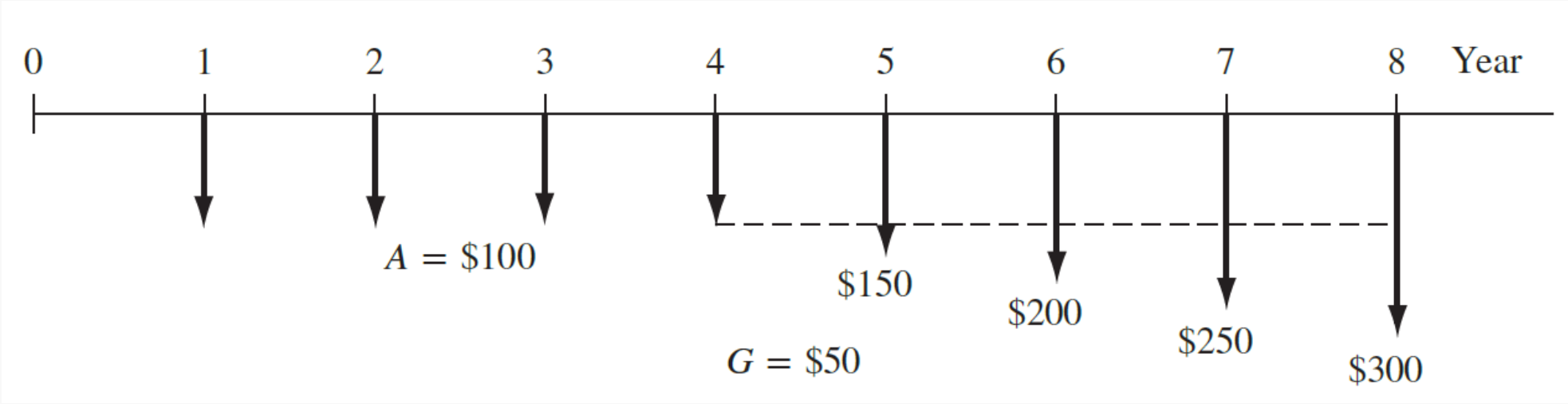
$$A_{1-10} = P_T(A/P,16\%,10) = 93{,}625(0.20690) = \$19{,}371$$

- **5. *A* over years 11 to 22:** For the equivalent 12-year series for years 11 through 22 (call it A_{11-22}), the *F value must be used* with the *A/F* factor for 12 years.

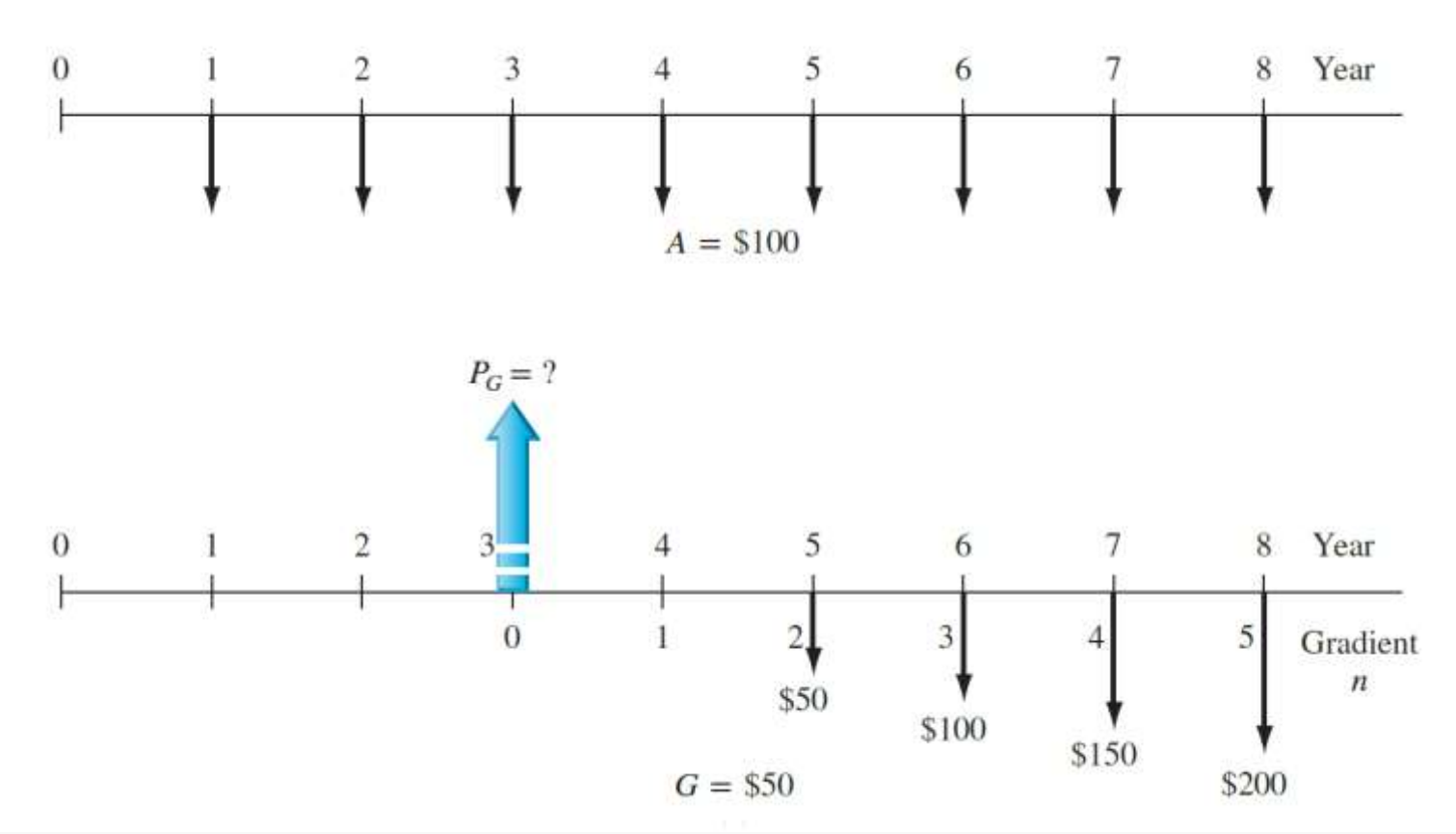
$$A_{11-22} = F(A/F,16\%,12) = 2{,}451{,}626(0.03241) = \$79{,}457$$

EXAMPLE

- Fujitsu, Inc. has tracked the average inspection cost on a robotics manufacturing line for 8 years. Cost averages were steady at $100 per completed unit for the first 4 years, but have increased consistently by $50 per unit for each of the last 4 years.
- Analyze the gradient increase, using the *P/G* factor. Where is the present worth located for the gradient?
- What is the general relation used to calculate total present worth in year 0?



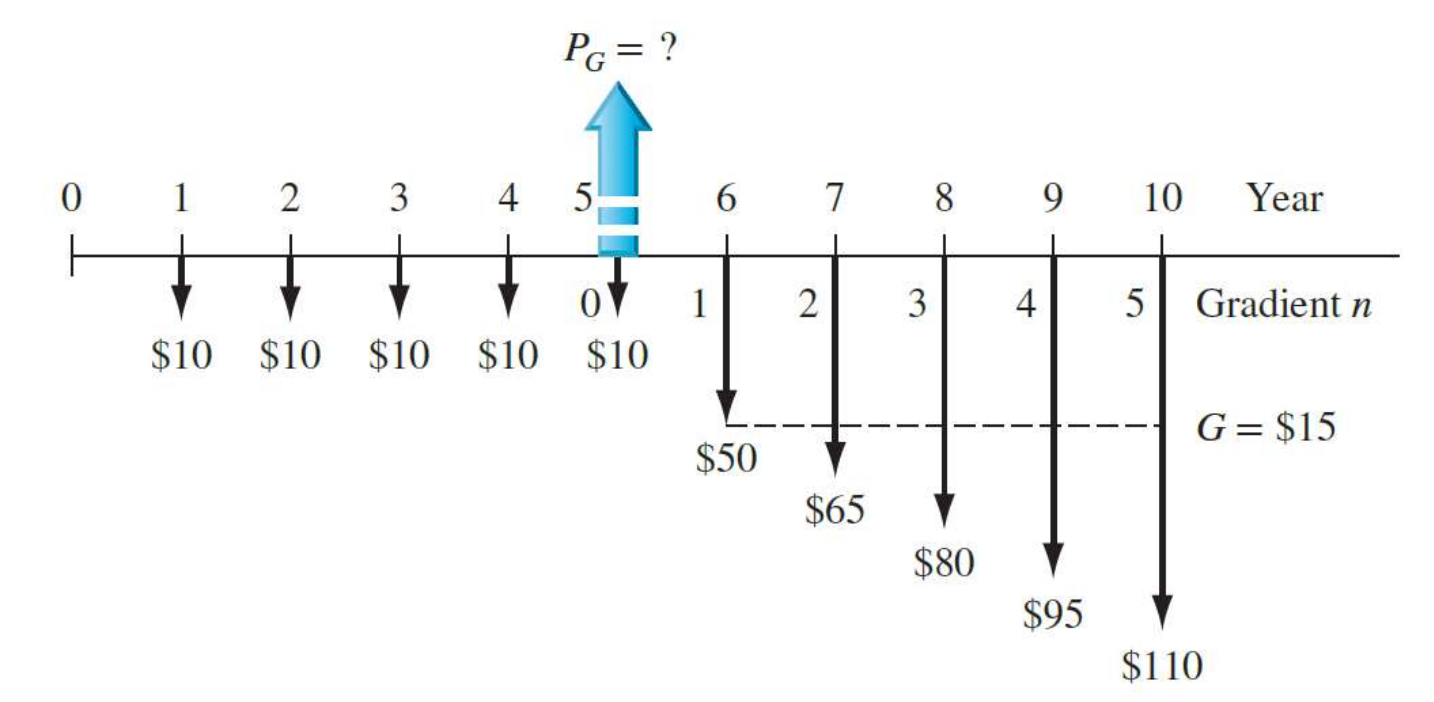
- The cash flow diagram shows the base amount A = \$100 and the arithmetic gradient G = \$50 starting between periods 4 and 5.
- Figure below partitions these two series. Gradient year 2 is placed in actual year 5 of the entire sequence. It is clear that n = 5 for the P/G factor. The P_G = ? arrow is correctly placed in gradient year 0, which is year 3 in the cash flow series.



- The uniform series A = \$100 occurs for all 8 years, and the G = \$50 gradient present worth PG appears in year 3.

$$P_T = P_A + P_G = 100(P/A,i,8) + 50(P/G,i,5)(P/F,i,3)$$

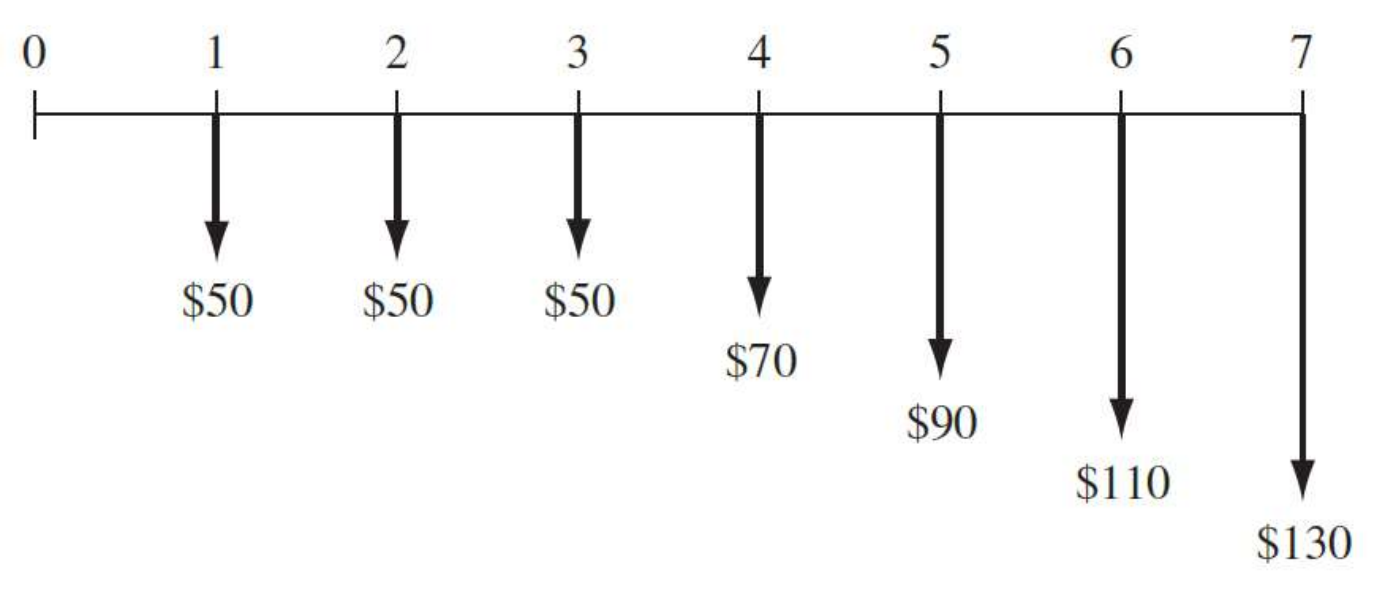
To find the equivalent A series of a shifted gradient through all the n periods, first find the present worth of the gradient at actual time 0, then apply the $(A/P,i,n)$ factor.



- To find the equivalent annual series in years 1 through 10 for the gradient series only, first find the present worth *PG* of the gradient in actual year 5, take this present worth back to year 0, and annualize the present worth for 10 years with the *A*/*P* factor. If you apply the annual series gradient factor (*A*/*G*,*i*,5) directly, the gradient is converted into an equivalent annual series over years 6 through 10 only, not years 1 through 10, as requested.

EXAMPLE

- Set up the engineering economy relations to compute the equivalent annual series in years 1 through 7 for the cash flow estimates in Figure below



The base amount annual series is $A_B = \$50$ for all 7 years Find the present worth P_G in actual year 2 of the \$20 gradient that starts in actual year 4. The gradient n is 5.

$$P_G = 20(P/G,i,5)$$

Bring the gradient present worth back to actual year 0.

$$P_0 = P_G(P/F,i,2) = 20(P/G,i,5)(P/F,i,2)$$

Annualize the gradient present worth from year 1 through year 7 to obtain A_G.

$$A_G = P_0(A/P,i,7)$$

Finally, add the base amount to the gradient annual series.

$$A_T = 20(P/G,i,5)(P/F,i,2)(A/P,i,7) + 50$$

Nominal and Effective Interest Rates

Nominal Interest Rate, r

Interest rate that does not include any consideration of compounding.

r = interest rate per period x number of periods

- A nominal rate r can be stated for any time period.
- Nominal rate of $r = 2\%$ per month is equal to
  - $2\% * 12$ months = 24% per year
  - $2\% * 3$ months = 6% per quarter
  - $2\% * 0.231$ month = 0.462% per week

- Effective interest rate is the actual rate that applies for a stated period of time. (The actual amount of interest that will accumulate in a stated period of time.)
- The compounding of interest during the time period of the corresponding nominal rate is accounted for by the effective interest rate.
- Effective interest rate is commonly expressed on an annual basis as the effective rate, i_a, but any time basis can be used.
- Effective rate has the compounding frequency attached to the nominal rate statement (4% per year, compounded monthly)
- If compounding frequency is not stated, it is assumed to be the same as the time period of *r* , i.e. compounding 1 time during the time period
- All the interest formulas, factors, tabulated values must have the effective interest rate to properly account for the time value of money.
- Therefore, we should know how to calculate the effective interest rate value for any nominal or effective rate statement.

- Annual Percentage Rate (APR) and Annual Percentage Yield (APY) can be used as instead of nominal and effective interest rates in practice.
- There are always three time-based units associated with an interest rate statement.
- Time Period- The period over which the interest is expressed. It is the *t* in the statement of *r% per time period t* (1% per month)
- Compounding Period- the shortest time period over which interest is charged or earned. (8% per year compounded monthly, therefore CP is month)
- Compounding frequency- the number of times that *m* compounding occurs within the time period *t*. If the compounding period (CP) and the time period *t* are the same, the compounding frequency is 1 (1% per month compounded monthly.)

Example

Consider the rate 8% per year compounded monthly.

- Time period = 1 year
- Compounding period = 1 Month
- Compounding frequency = 12 times per year

- So far all the interest rates used have *t* and *m* as 1 year (8% per year). The effective and nominal rates were the same. It is common practice to express the rates on the same time basis as the compounding period.

$$\text{Effective rate per CP} = \frac{r\% \text{ per time period } t}{m \text{ compounding periods per } t} = \frac{r}{m}$$

Example

Assume $r = 9\%$ per year, compounded monthly, then $m = 12$ and the effective rate per the compounding period is found as $\frac{9\%}{12} = 0.75\%$ per month, compounded monthly.

- NOTE: Changing the basic time period t does not alter the compounding period, month in the previous example.
- Whenever time period = compounding period, stated nominal rate is an effective rate. Therefore, i_{cp} is the effective rate per CP too.

Statement	CP	What This Is
$i = 10\%$ per year	CP not stated; CP = year	Effective rate per year
$i =$ effective 10% per year, compounded monthly	CP stated; CP = month	Effective rate per year
$i = 1\frac{1}{2}\%$ per month	CP not stated; CP = month	Effective rate per month
$i =$ effective $1\frac{1}{2}\%$ per month, compounded monthly	CP stated; CP = month	Effective rate per month; terms *effective* and *compounded monthly* are redundant
$i =$ effective 3% per quarter, compounded daily	CP stated; CP = day	Effective rate per quarter

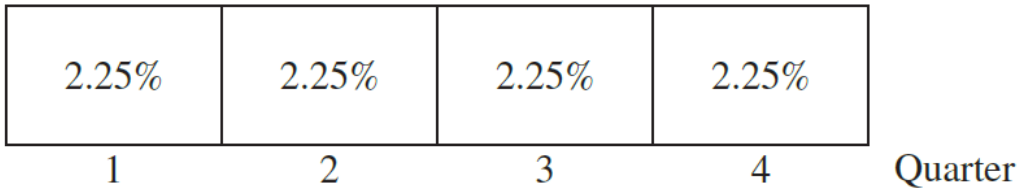
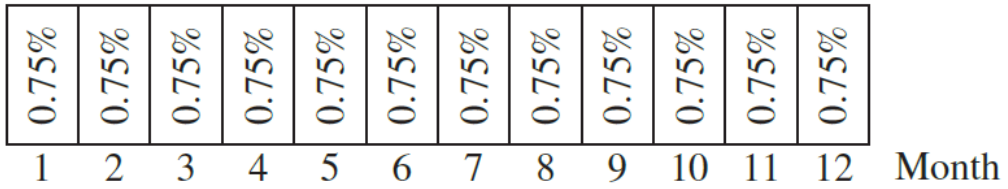
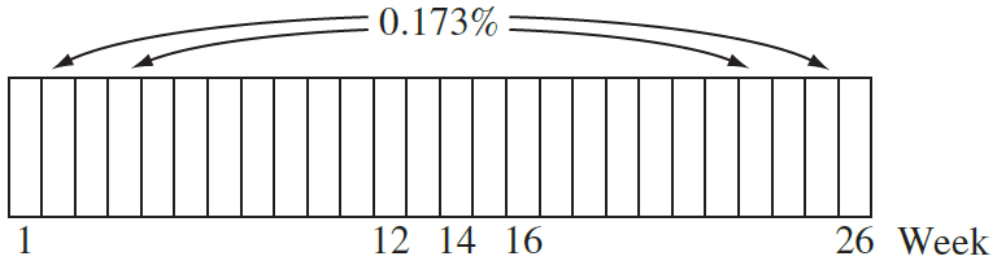
EXAMPLE

- Three different bank loan rates for electric generation equipment are listed below. Determine the effective rate on the basis of the compounding period for each rate.

  (*a*) 9% per year, compounded quarterly.

  (*b*) 9% per year, compounded monthly.

  (*c*) 4.5% per 6 months, compounded weekly.

Nominal r% per t	Compounding Period (CP)	Compounding Frequency (m)	Rate per CP $\left(\frac{r}{m}\right)$	Distribution over Time Period t
(a) 9% per year	Quarter	4	2.25%	
(b) 9% per year	Month	12	0.75%	
(c) 4.5% per 6 months	Week	26	0.173%	

The Credit Card Offer Case:

- Today, Dave received a special offer of a new credit card from Chase Bank linked with the majör airline that he flies frequently.
- It offers a generous bonus package for signing up by a specific date about 60 days from now.
- The bonus package includes extra airline points (once the first purchase is made), priority airport check-in services (for 1 year), several free checked-bag allowances (for up to 10 check-ins), extra frequent-flyer points on the airline, Access to airline lounges (provided he uses the card on a set time basis), plus several other rewards (rental car discounts, cruise trip amenities, and floral order discounts).

- The annual fee of $85 for membership does not start until the second year, and balance transfers from other credit cards have a low transfer fee, provided they are made at the time of initial membership.
- Dave has a credit card currently with a bank that he is planning to leave due to its poor customer service and high monthly fees.
- If he enrolls, he will transfer the $1000 balance on the current card to the new Chase Bank card. I
- n the page that accompanies the offer letter, “pricing information” is included. This includes interest rates, interest charges, and fees. A summary of several of these rates and fees follows.

■ APR (annual percentage rate) for purchases and balance transfers	■ **14.24% per year** (sum of the current U.S. Government prime rate of 3.25% and 10.99%, which is the APR added to determine the balance transfer APR for Chase Bank)
■ APR for cash and overdraft advances	■ **19.24% per year**
■ Penalty APR for late minimum payment, exceeding credit limit, and returned unpaid payments	■ **29.99% per year (maximum penalty APR)**

- **Fees are listed as follows:**

Annual membership	$85; free the first year
Balance transfers	$5 or 3% of each transfer, whichever is greater
Cash advances	$10 or 3% of each advance, whichever is greater
Late payment	$39 each occurrence, if balance exceeds $250
Over the credit limit	$39 each occurrence
Returned check or payment	$39 each occurrence

- All APR rates are variable, based on a current 3.25% prime rate with 10.99% added to determine purchase/balance transfer APR; with 15.99% added to determine cash/overdraft APR; and with 26.99% added to determine penalty APR. †The penalty APR applies indefinitely to future transactions. If no minimum payment is received within 60 days, the penalty APR applies to all outstanding balances and all future transactions on the account.

- The balance transfer APR interest rate of 14.24% is an annual rate, with no compounding period mentioned. Therefore, we should conclude that the CP is 1 year, the same as the annual interest period of the APR. However, as Dave and we all know, credit card payments are required monthly.
- (*a*) Determine the effective interest rates for *compounding periods of 1 year and 1 month* so Dave knows some effective rates he might be paying when he transfers the $1000 balance from his current card.
- (*b*) Assume that immediately after he accepts the card and completes the $1000 transfer, Dave gets a bill that is due 1 month later. What is the amount of the total balance he owes?
- Now, Dave looks a little closer at the fine print of the “pricing information” sheet and discovers a small-print statement that Chase Bank uses the daily balance method (including new transactions) to determine the balance used to calculate the interest due at payment time.
- (*c*) We will reserve the implication of this new finding until later, but for now help Dave by determining the *effective daily interest rate* that may be used to calculate interest due at the end of 1 month, provided the CP is 1 day.

- (*a*) The interest period is 1 year

  CP of year: Effective rate per year = 14.24⁄1 = 14.24%

  CP of month: Effective rate per month = 14.24⁄12 = 1.187%

- (*b*) The interest will be at the monthly effective rate, plus the balance transfer fee of 3%.

$$\text{Amount owed after 1 month} = 1000 + 1000(0.01187) + 0.03(1000)$$
$$= 1000 + 11.87 + 30$$
$$= \$1041.87$$

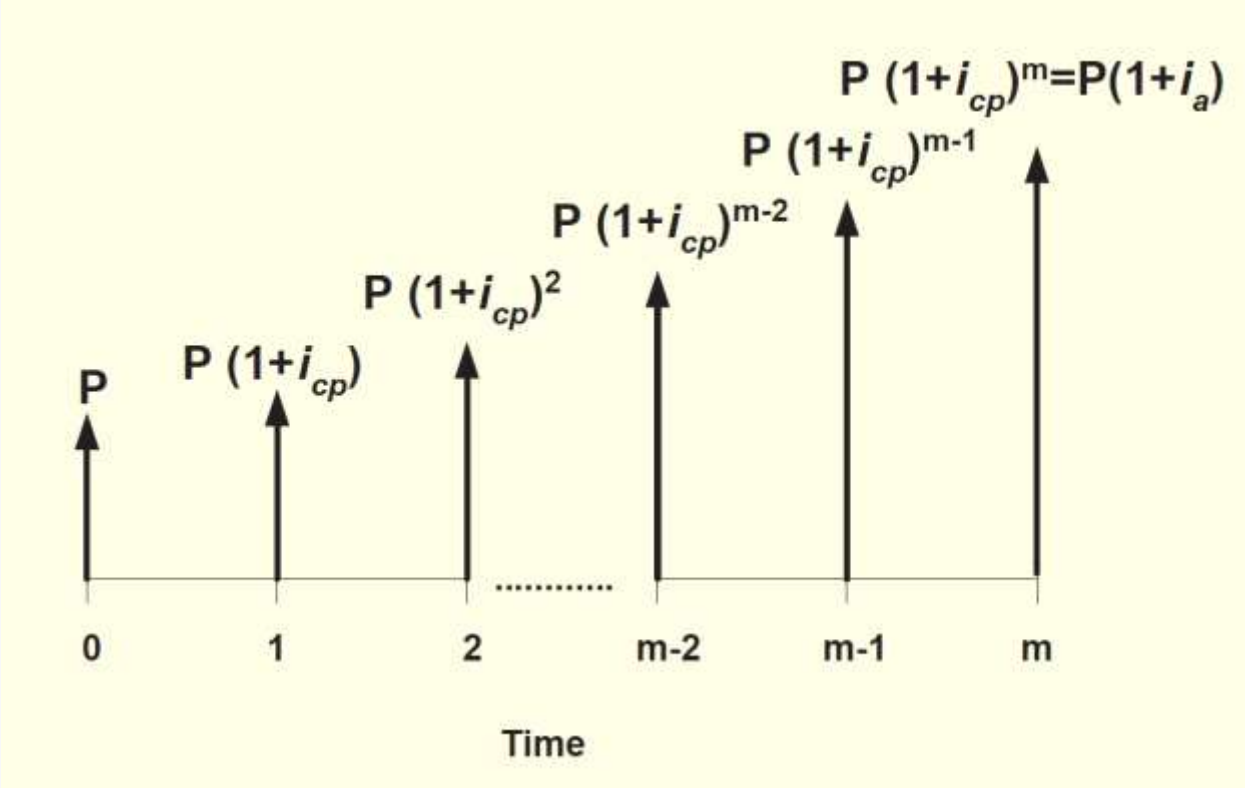
Including the $30 fee, this represents an interest rate of (41.87⁄1000)(100%) = 4.187% for only the 1-month period.

- (*c*) now $m = 365$ compounding periods per year.

  CP of day: Effective rate per day = 14.24⁄365 = 0.039%

- For the Effective Annual Interest rate, time period t is year, and the compounding period can be any time unit less than a year.
- Effective interest rate at any point during the year includes the interest rate of all previous compounding periods during the year.
- Derivation of the effective annual interest rate formula directly parallels the logic of the future worth relation $F = P(1 + i)n$.
- Rewrite the formula as $F = P(1 + i_a)$, since interest can be compounded many times a year.
- The rate i_{cp} per CP must be compounded through all m years to find the total effect of compounding by the end of year.

- r : nominal interest rate per year
- m: compounding occurs within the time period t (1 year).
- i_{cp}: effective interest rate per compounding period (r/m)
- i_a: effective interest rate per year



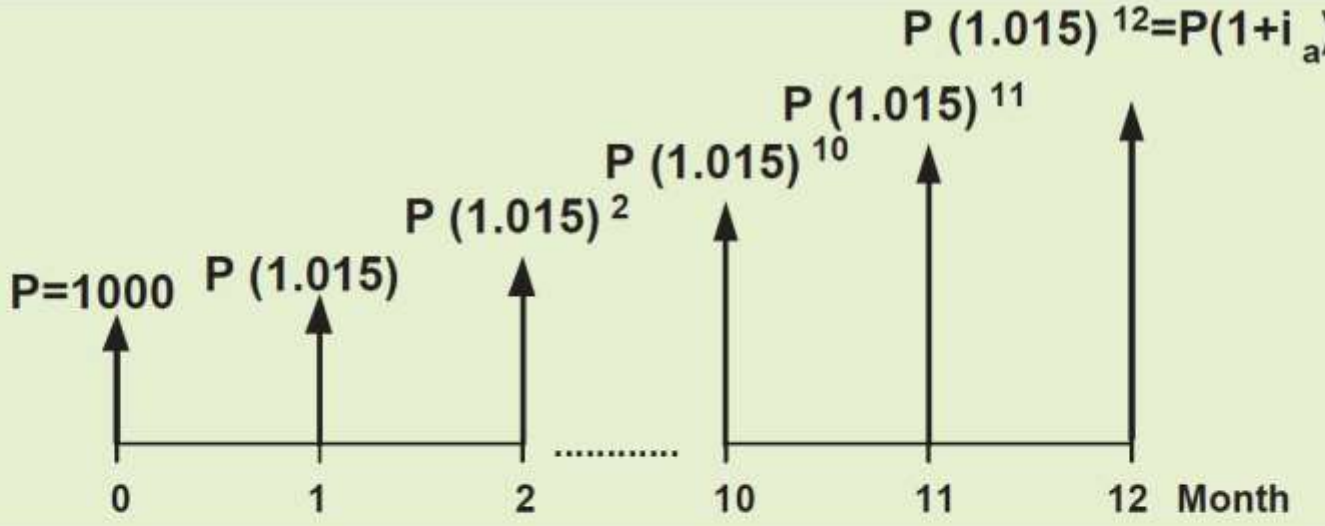
$$
\begin{aligned}
(1 + i_a) &= (1 + i_{cp})^m \\
i_a &= (1 + i_{cp})^m - 1 \\
i_a &= (1 + r/m)^m - 1
\end{aligned}
$$

EXAMPLE

- Kelso obtained a new credit card from a national bank, MBNA, with a stated rate of 18% per year, compounded monthly. For a $1,000 balance at the beginning of the year, find the effective annual rate and the total amount owed to MBNA after 1 year, provided no payments are made during the year.

  - r (nominal annual rate)=18%, m=12,

  - i_{cp} (interest per CP)=$\frac{r}{m} = \frac{18\%}{12} = 1.5\%$ per month

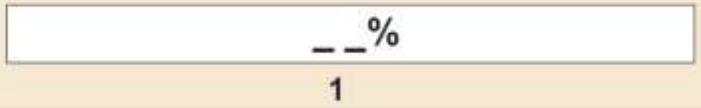
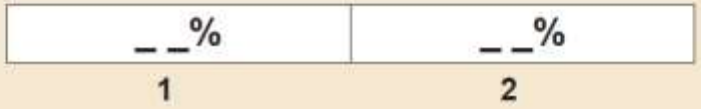
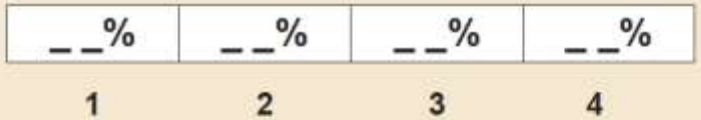
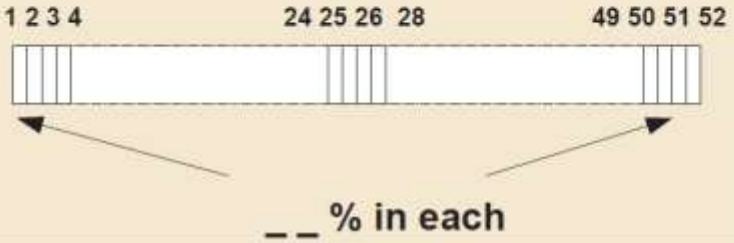


- $i_a = (1 + i_{cp})^{12} - 1 = (1.015)^{12} - 1 = 0.19562$

- $F = 1000(1 + i_a) = 1000(1.19562) = \$1,195.62$

In Class Work 4

Calculate the effective annual interest rates when $r = 18\%$ per year, compounded m-ly

Compounding Period	Rate over CP, i_{cp}	m	Effective Annual Rate i_a
Year			
Semi Annual			
Quarter			
Month			
Weekly			

Compounding Period	Rate over CP, i_{cp}	m	Effective Annual Rate i_a
Year	18% 1	1	$(1.18)^1-1= 18\ \%$
Semi Annual	9% (1) 9% (2)	2	$(1.09)^2-1= 18.81\ \%$
Quarter	4.5% (1) 4.5% (2) 4.5% (3) 4.5% (4)	4	$(1.045)^4-1= 19.252\ \%$
Month	1 2 3 4 5 6 7 8 9 10 11 12 1.5% in each	12	$(1.015)^{12}-1= 19.562\ \%$
Weekly	1 2 3 4 ... 24 25 26 28 ... 49 50 51 52 0.34615 % in each	52	$(1.0034615)^{52}-1= 19.684\%$

- Now, we need to introduce the payment period, the frequency of payments or receipts. Most of the time the compounding period and payment period are not the same.
- A company deposits money into an account each month that pays a nominal interest rate of 12% per year, compounded semiannually.
  - Compounding period = 6 months
  - Payment period = 1 month
- To evaluate cash flows occurring more frequently than annually, effective interest rate over the payment period must be calculated.
- Let *r* : nominal interest rate per payment period
- Let *m*: number of compounding periods per payment period
- Effective interest:

$$i = \left(1 + \frac{r}{m}\right)^m - 1$$

Meaning of the Effective Interest Rate (EIR)

- The **Effective Interest Rate (EIR)** — sometimes called the **Effective Annual Rate (EAR)** — tells you the **true annual return or cost** of money **after accounting for compounding.**
- In other words:
- It shows how much your money actually grows (or how much you really pay) in one year, given that interest is added to the principal more than once a year.

- **Why do we need it?**
- Banks, loans, and investments often quote a **nominal rate (r)** — like *12% per year, compounded monthly*.
  That means interest is applied **12 times a year**, not just once.
- However, because interest earns more interest each period (compounding), the **actual annual return** is **higher than 12%**.
- So the **effective rate (i)** translates that compounding effect into **a single, once-per-year rate** — the *true equivalent*.

- Imagine two investments:

Investment	Quoted rate	Compounding	Actual gain after 1 year
A	12% per year (annual)	Once per year	12.00%
B	12% per year (monthly)	12 times per year	12.68%

- Both *sound* like 12%,
- but B actually gives you **more money** because it compounds interest **every month**.

EXAMPLE

- A bank offers a **nominal annual interest rate** of

  r=12% per year, compounded monthly.

- Find the **effective annual interest rate i.**

$$i = \left(1 + \frac{r}{m}\right)^m - 1$$

where:

- r = nominal annual rate
- m = number of compounding periods per year

$$r = 0.12, \quad m = 12$$

$$i = \left(1 + \frac{0.12}{12}\right)^{12} - 1$$

$$i = (1.01)^{12} - 1 = 1.126825 - 1 = 0.126825$$

$$\boxed{i = 12.68\%}$$

- Although the nominal rate r is 12%, the **effective annual rate** i is **12.68%**, because interest is compounded monthly.

Comparison of Compounding Frequencies

Compounding Frequency	m	Formula	Effective Rate i
Annual	1	$(1 + 0.12/1)^1 - 1$	12.00%
Semiannual	2	$(1 + 0.12/2)^2 - 1$	12.36%
Quarterly	4	$(1 + 0.12/4)^4 - 1$	12.55%
Monthly	12	$(1 + 0.12/12)^{12} - 1$	12.68%
Daily (365)	365	$(1 + 0.12/365)^{365} - 1$	12.75%

Example

Three suppliers place their bids for the procurement of a major part for a motor company. The motor company will make semiannual payments. The bids by the suppliers are given as:

1. 9% per year, compounded quarterly,
2. 9% per year, compounded monthly,
3. 3% per quarter, compounded quarterly.

Determine the effective rate for each bid on the basis of semiannual payments.

For bid:

1. $i = (1 + \frac{0.045}{2})^2 - 1 = (1.0225)^2 - 1 = 4.55\%$
2. $i = (1 + \frac{0.045}{6})^6 - 1 = (1.0075)^6 - 1 = 4.58\%$
3. $i = (1 + \frac{6}{2})^2 - 1 = (1.03)^2 - 1 = 6.09\%$

EXAMPLE

- Three suppliers place their bids for the procurement of a major part for a motor company. The motor company will make semiannual (every 6 months) payments. The bids by the suppliers are given as:

  a) 9% per year, compounded quarterly,

  b) 9% per year, compounded monthly,

  c) 3% per quarter, compounded quarterly.

  Determine the effective rate for each bid on the basis of semiannual payments.

$$i = \left(1 + \frac{r}{m}\right)^{m \times t} - 1$$

where:

- r = nominal annual rate
- m = number of compounding periods per year
- t = desired time in years (for semiannual, $t = 0.5$)

- 9% per year, compounded quarterly

$$r = 0.09,\ m = 4$$

$$i_1 = \left(1 + \frac{0.09}{4}\right)^{4\times 0.5} - 1$$

$$i_1 = (1.0225)^2 - 1 = 0.045506$$

$$\boxed{i_1 = 4.55\% \text{ per semiannual period.}}$$

- 9% per year, compounded monthly

$$r = 0.09,\ m = 12$$

$$i_2 = \left(1 + \frac{0.09}{12}\right)^{12\times 0.5} - 1$$

$$i_2 = (1.0075)^6 - 1 = 0.045677$$

$$\boxed{i_2 = 4.57\% \text{ per semiannual period.}}$$

- 3% per quarter, compounded quarterly

3% per **quarter**, compounded quarterly → quarterly effective rate = 0.03

$$i_3 = (1 + 0.03)^2 - 1 = 0.0609$$

$$\boxed{i_3 = 6.09\% \text{ per semiannual period.}}$$

Supplier	Interest Term	Semiannual Effective Rate i
1	$r = 9\%$, compounded quarterly	**4.55%**
2	$r = 9\%$, compounded monthly	**4.57%**
3	$r = 3\%$ per quarter	**6.09%**

- Therefore, **Supplier 1** provides the most favorable financing terms for the motor company.

- Allowing more and more frequent compounding, increases the number of compounding periods per time period, *m*. As *m* approaches infinity, the effective interest rate must be written in a new form.

$$lim_{h\to\infty}\left(1+\frac{1}{h}\right)^{h} = e = 2.71828$$

The limit of $i = (1+\frac{r}{m})^{m} - 1$ as *m* approaches infinity is found by setting $r/m = 1/h$, which makes $m = hr$.

$$\begin{aligned} lim_{m\to\infty} i &= lim_{m\to\infty}\left(1+\frac{r}{m}\right)^{m} - 1 \\ &= lim_{h\to\infty}\left[\left(1+\frac{1}{h}\right)^{h}\right]^{r} - 1 \\ i &= e^{r} - 1 \end{aligned}$$

- Therefore, **Supplier 1** provides the most favorable financing terms for the motor company.
- This formula compute the effective continuous interest rate when the time periods on *i* and *r* are the same.

Example

For an interest rate of 18% per year, compounded continuously, calculate the effective monthly and annual interest rates.

- Nominal monthly interest rate $r = 18\%/12 = 1.5\%$.
- Effective monthly rate $i\%$ per month $= e^{0.015} - 1 = 1.511\%$
- Nominal annual interest rate $r = 18\%$.
- Effective annual rate $i\% = e^{0.18} - 1 = 19.72\%$

An investor requires an effective return of at least 15%. What is the minimum annual nominal rate that is acceptable for continuous compounding.

$$\begin{aligned} 0.15 &= e^r - 1 \\ 1.15 &= e^r \\ \ln(1.15) &= \ln e^r \\ r &= 0.1397 \rightarrow 13.97\% \end{aligned}$$